

CME 222

MASS TRANSFER



Introduction

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Moodle Enrolment

Course: CME222 Mass Transfer 2026/04 Peng Chee Tan

Enrolment key: CME222

Course Learning Outcome

- Apply the fundamental knowledge of mass transfer in practical problem
- Distinguish the mechanisms of mass transfer and calculate the mass transfer diffusion coefficients of gases, liquids, and solids
- Analyse diffusional process and calculate the flux in one phase and two-phase system

Course Learning Outcomes	Programme Learning Outcomes (PLO)											
	Engineering Knowledge	Problem Analysis	Design/ Development of Solutions	Investigation	Modern Tool Usage	The Engineer and Society	Environmental and Sustainability	Ethics	Communication	Individual and Team Work	Life Long Learning	Project Management and Finance
	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	✓											
CLO2		✓										
CLO3		✓										

Course Structure

Credit Value

- 3 credits (2h lecture +2h tutorial)

Lecture & Tutorial Class

- Lecture: Wednesday 1 pm to 3 pm
- Tutorial: Thursday 1 pm to 3 pm (Group 1)
Thursday 3 pm to 5 pm (Group 2)

Please be punctual!

Course Structure

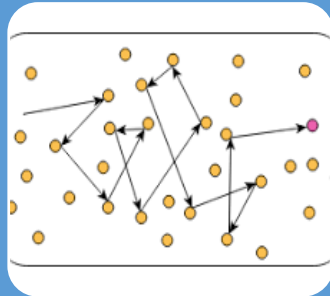
Assessment

Assessment	Weightage (%)
Assignment	25
Test (Week 8)	25
Final Examination	50
TOTAL	100

Passing Criteria

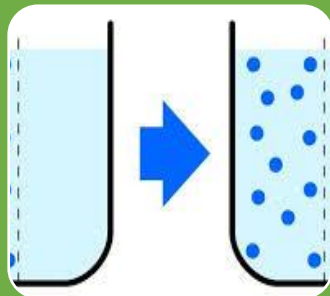
- Pass **Final** Exam (score **above 40**) **AND**
- **Total** mark **above 55**

Course Content



Introduction

- Basic concept of mass transfer
- Momentum, energy and mass transfer analogies



Diffusion

- Concentration, velocities, fluxes
- Molecular diffusivity
- Fick's first law, Fick's second law
- Equimolar counter diffusion, Unimolecular diffusion
- Transient diffusion

Course Content



Convective Mass Transfer

- Mass transfer coefficient for various geometry (plate, spheres, cylinder and etc.)
- Laminar concentration boundary layer



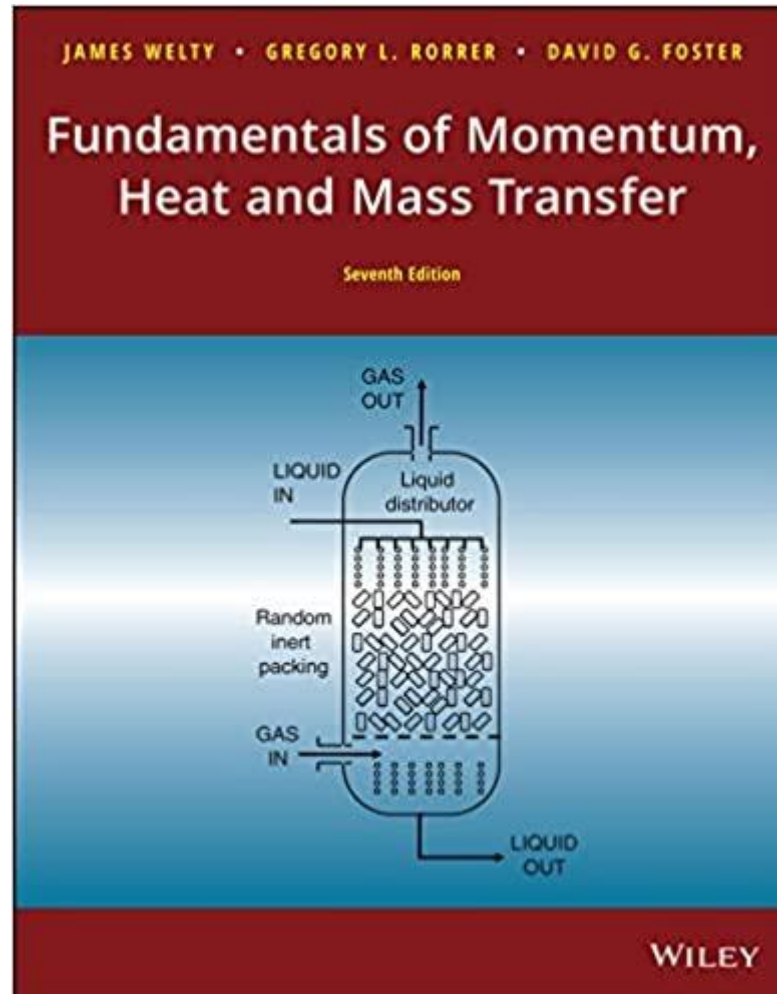
Interphase Mass Transfer

- Diffusion between phases
- Two-resistance film theory
- Gas-liquid absorption

References

- James Welty, Gregory L. Rorrer, David G. Foster (2019). Fundamentals of Momentum, Heat and Mass Transfer. 7th Edition, Wiley
- Theodore L. Bergman, Adrienne S. Lavine, Frank P. Incropera, David P. DeWitt (2016). Fundamentals of Heat and Mass Transfer 8th Edition. Wiley.
- R. Byron Bird, Warren E. Stewart, Edwin N. Lightfoot, Daniel J. Klingenberg (2014). Introductory Transport Phenomena, 1st Edition, Wiley.

References



Consultation Hours

Tuesday

10 am to 12 pm

Tuesday

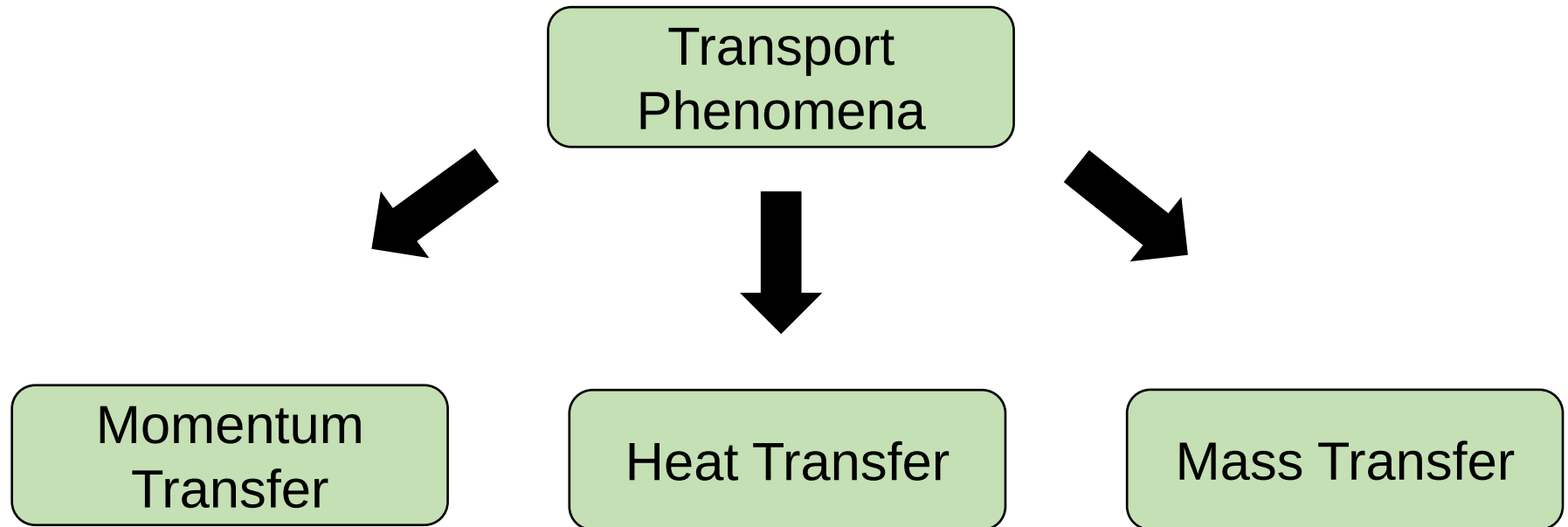
2 pm to 4 pm

Table of Content

- Basic Concept of Mass Transfer
- Momentum, Energy, and Mass Transfer Analogies

Basic Concepts of Mass Transfer

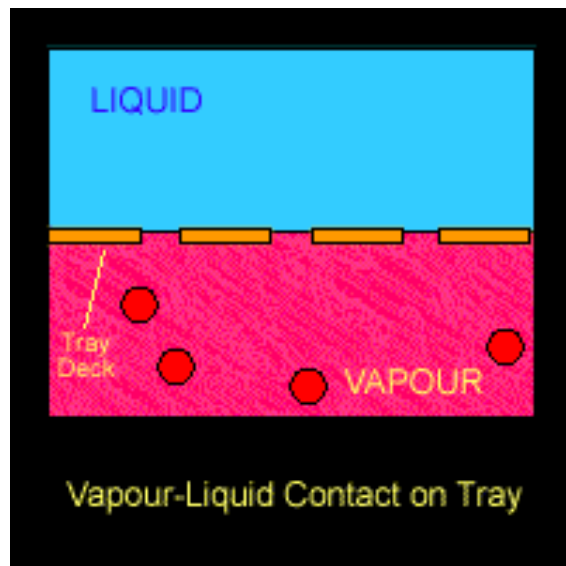
Transport Phenomena



Transport Phenomena

These three transports should be studied together:

- Frequently occur simultaneously in industrial process
- The basic equations are closely related



What is Mass Transfer??

- Mass transfer is defined as the transports of one or more chemical species in fluid mixture due to some driving force
- Driving force: concentration differences, temperature differences, pressure differences, difference due to gravity, magnetic or electrical field

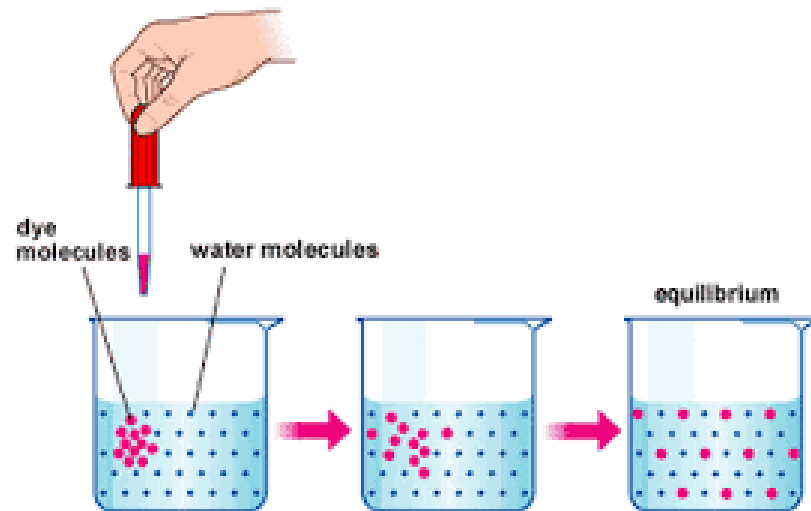
What is Mass Transfer??

Concentration Gradient

There is a natural tendency for mass to transfer in a system with two or more components when their concentrations vary from point to point, minimizing these differences



Evaporation

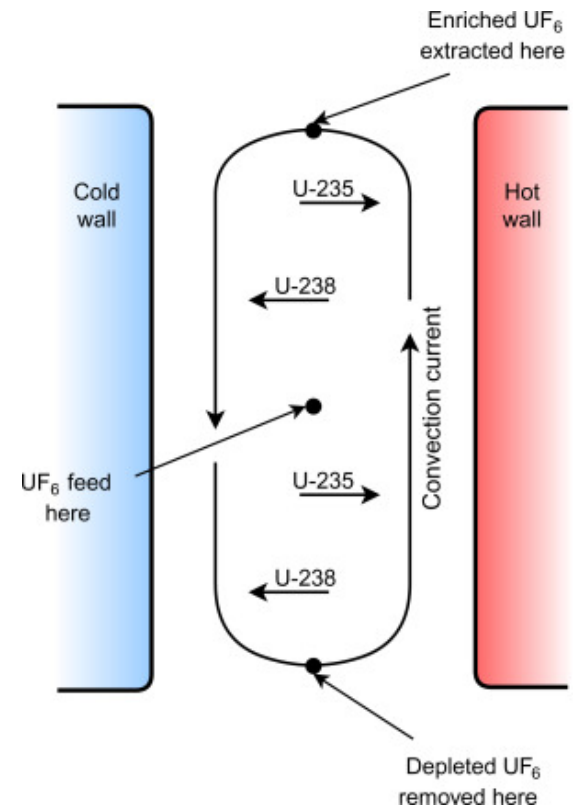


Dye dispersion

What is Mass Transfer??

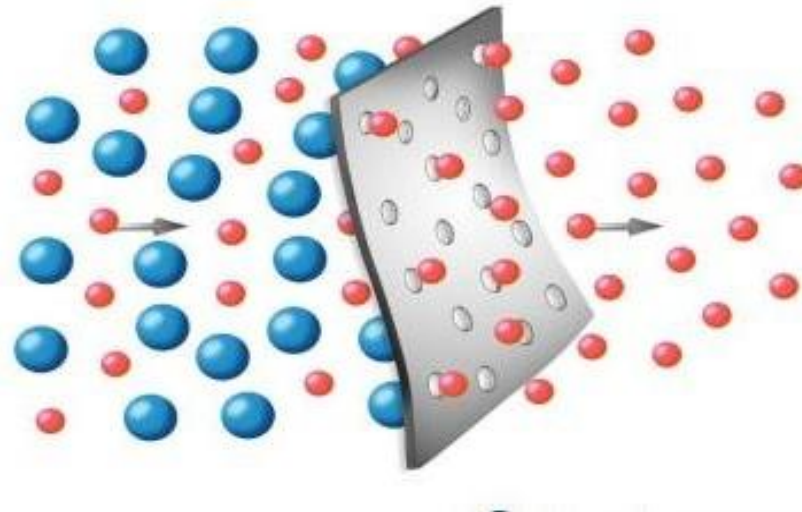
Temperature Gradient

- Separation of uranium isotopes
- Liquid uranium hexafluoride feed into the space between two concentric vertical pipes
- Upward flow on hot side, downward flow on cold side
- Heavier U-238 accumulate near cold side
- Lighter accumulate near hot side
- Taller column $\rightarrow$ more efficiency separation



What is Mass Transfer??

Pressure difference

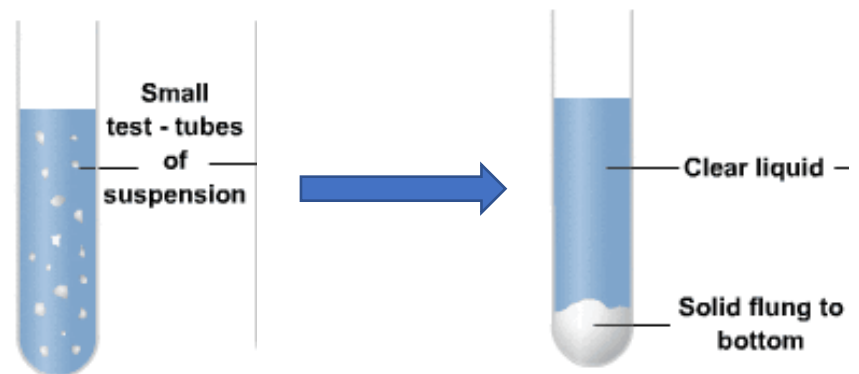
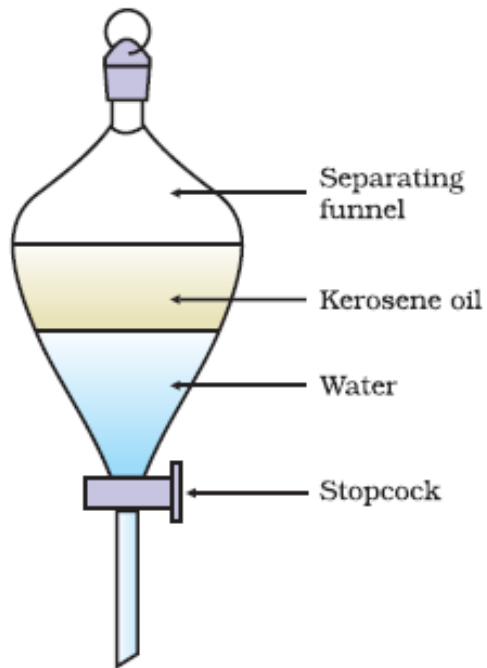


Membrane

What is Mass Transfer??

Gravity Effect

- Due to difference in density

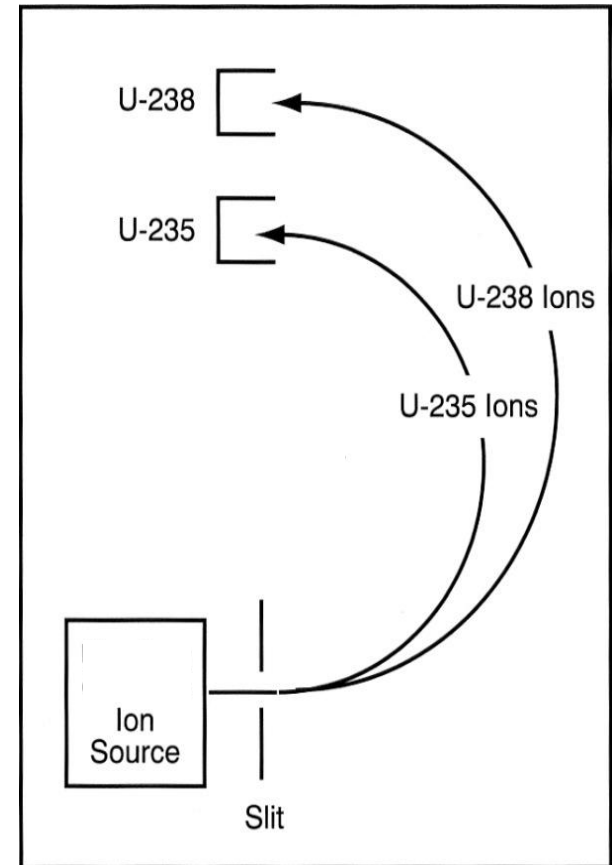


Centrifugation

What is Mass Transfer??

Magnetic Field

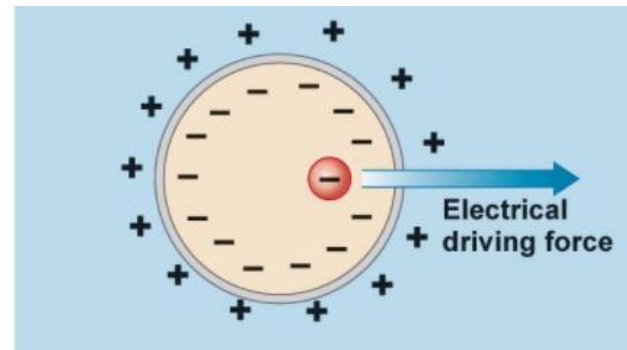
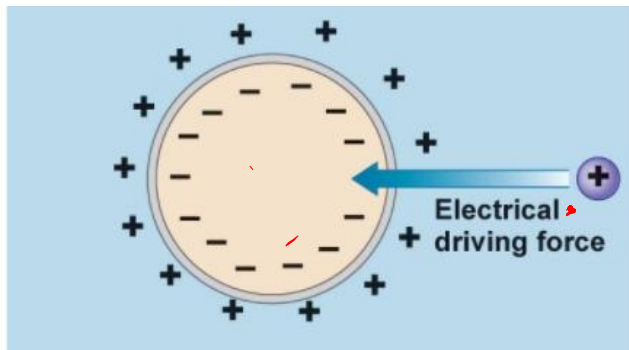
- Charged particle send through magnetic field
- Charged particle experiences a force when moving through a magnetic field
- Ion deflected



What is Mass Transfer??

Electric Field

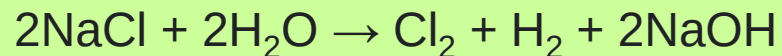
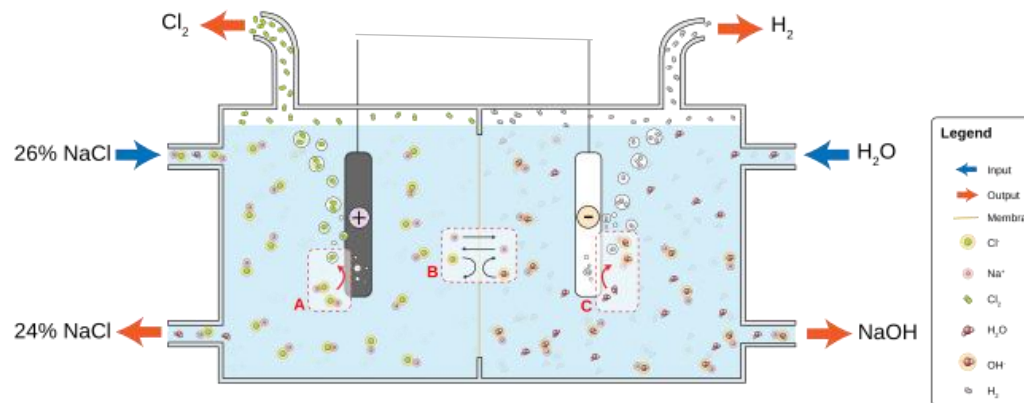
- Ions, atoms or molecules that have a charge, can be affected by an **electrical driving force**
- The direction of the electrical driving force results in the ion moving toward a region where the opposite charge exists



What is Mass Transfer??

Electric Field

- Cl^- is oxidized to chlorine at anode (A)
- Selective membrane allows Na^+ to freely flow across
- At cathode, water is reduced to hydroxide and hydrogen gas

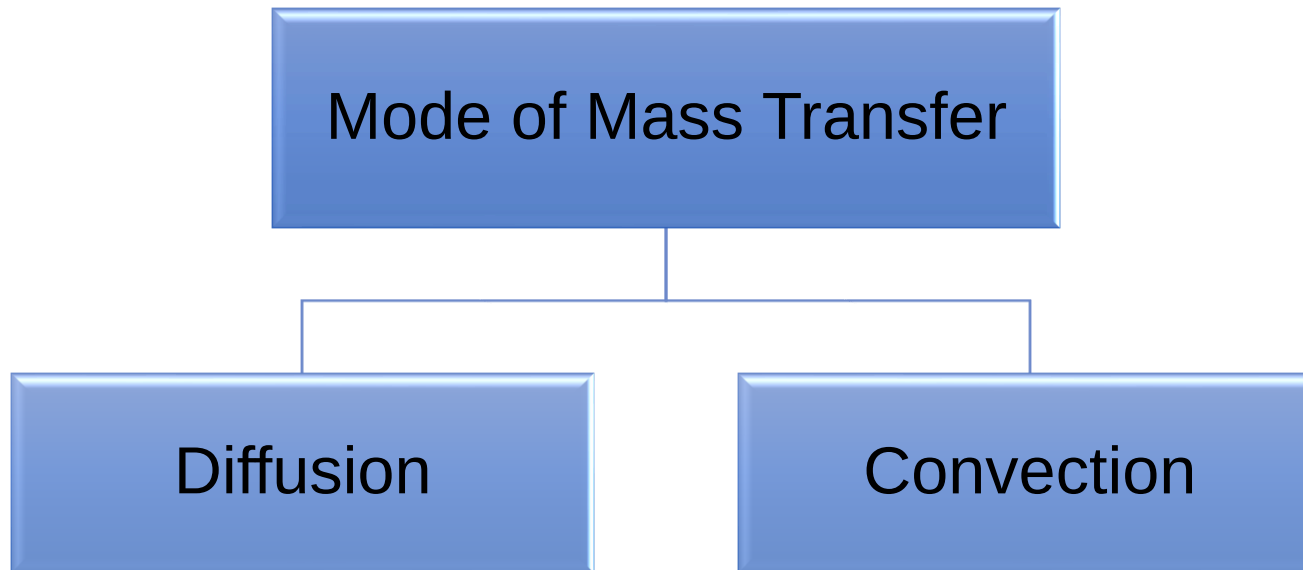


More Examples...

- Perfume presents a pleasant fragrance that is imparted throughout the surrounding atmosphere
- Drying of clothes
- Evaporation of water spilled on the floor when water molecules diffuse into the air surrounding it
- Dispersion of air pollutants released from a power plant smoke stack into the environment



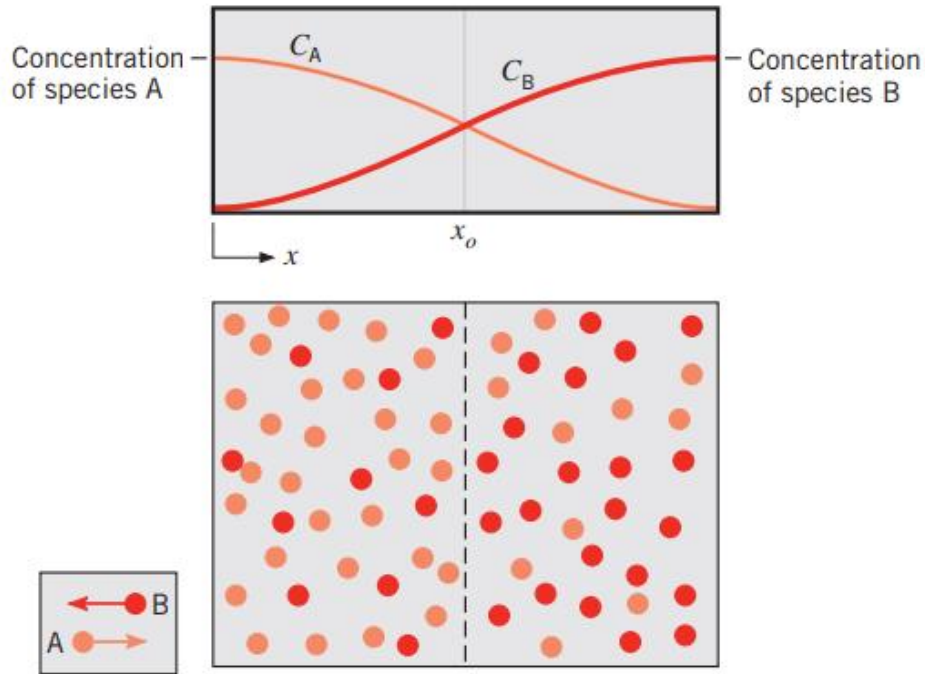
Mode of Mass Transfer



Diffusion

- Diffusion is the transport of **molecules, ions, or small particles** from a region of relatively **high concentration** into a region of **lower** concentration
- In the absence of a concentration gradient, molecules continue to move randomly (**Brownian motion**), but no net mass transfer occurs and the system remains uniform

Diffusion

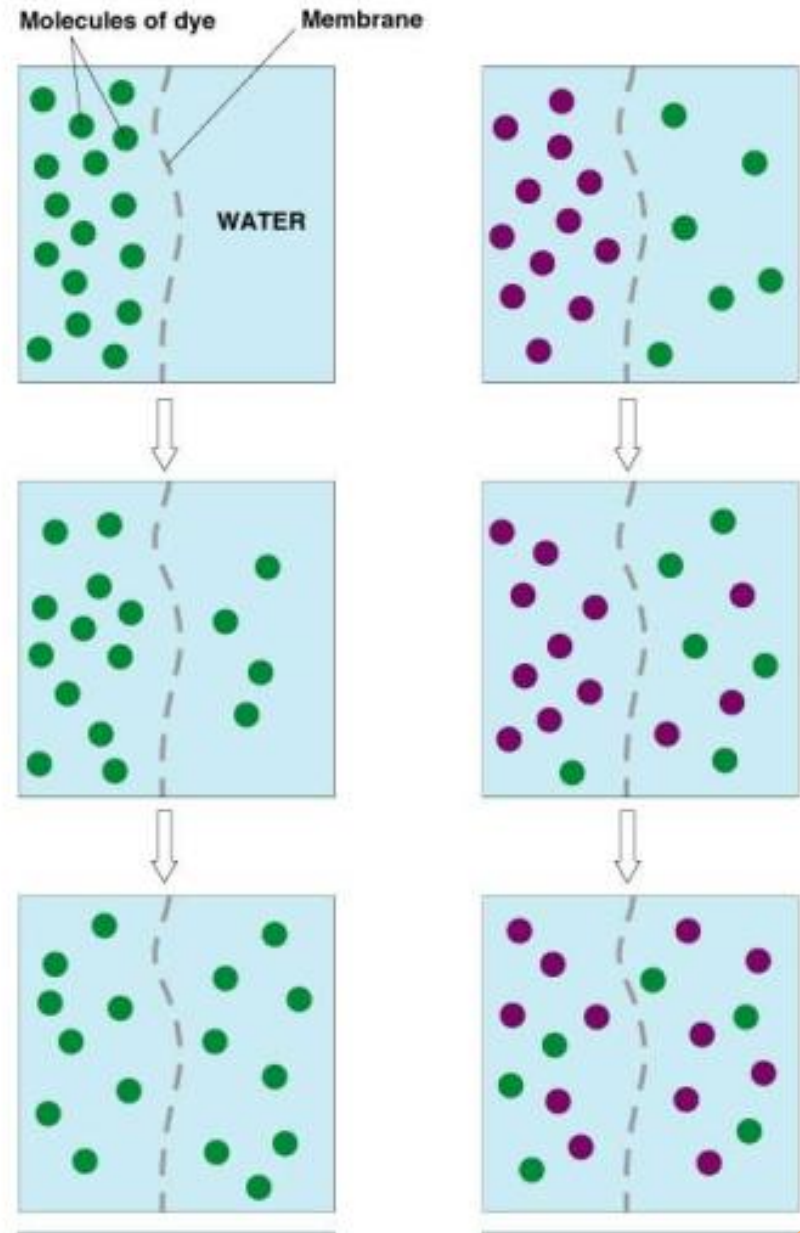


- Mass diffusion is in the direction of decreasing concentration
- After a sufficient time, **uniform concentrations** of A and B are achieved
- There is no net transport of species A or B across the imaginary plane

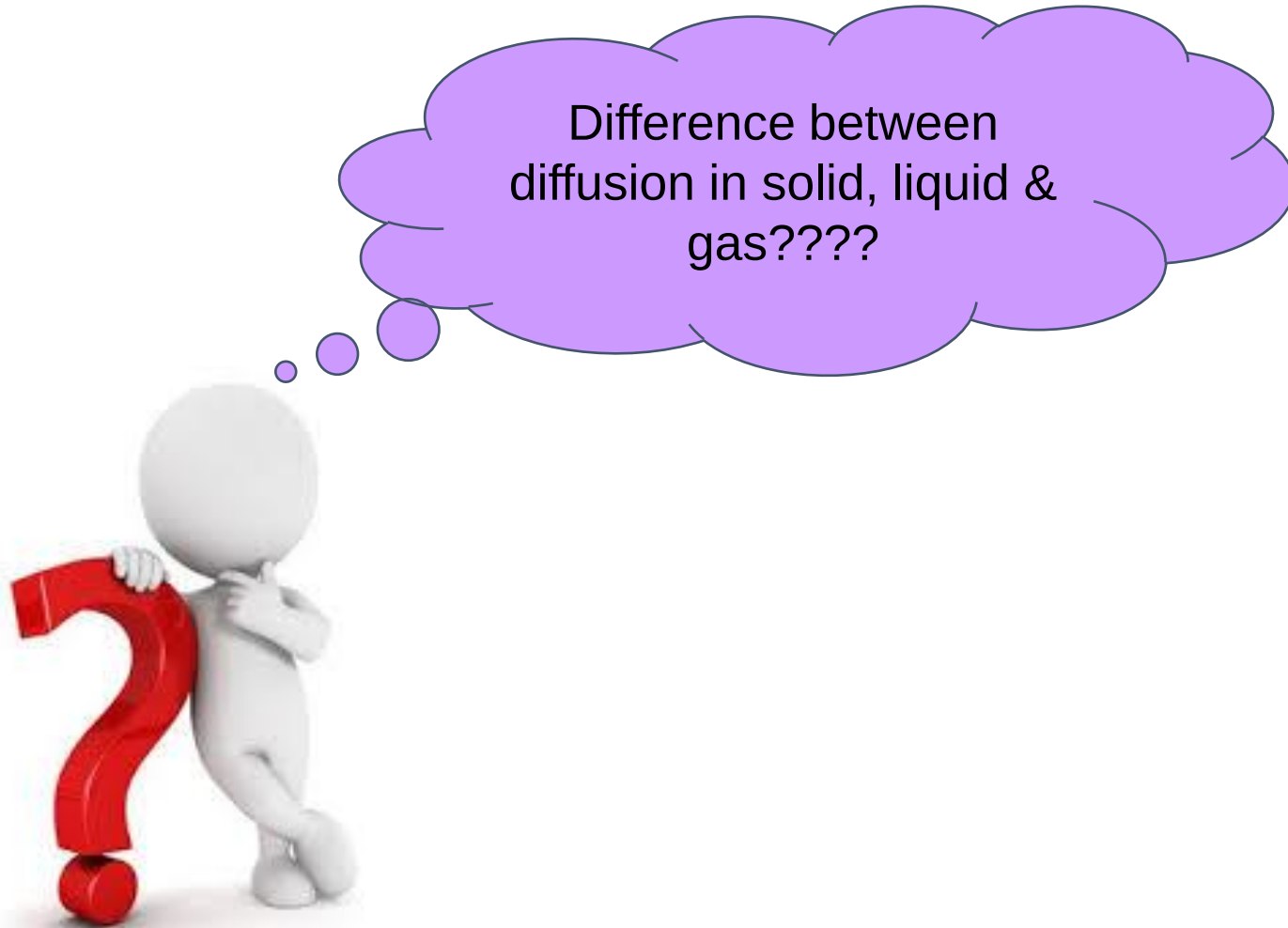
Diffusion



Diffusion stops when reaches equilibrium

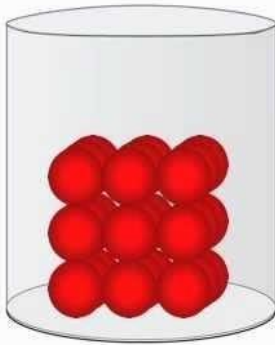


Diffusion



Diffusion

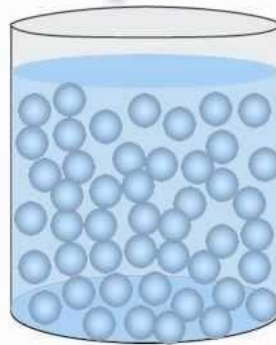
solid



- rigid
- fixed shape
- fixed volume

cannot be squashed

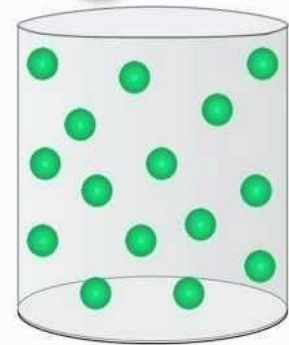
liquid



- not rigid
- no fixed shape
- fixed volume

cannot be squashed

gas



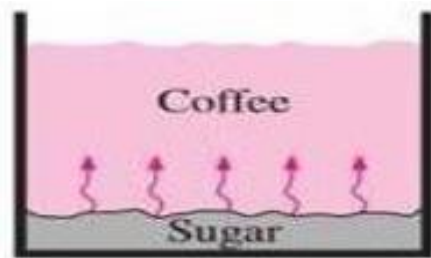
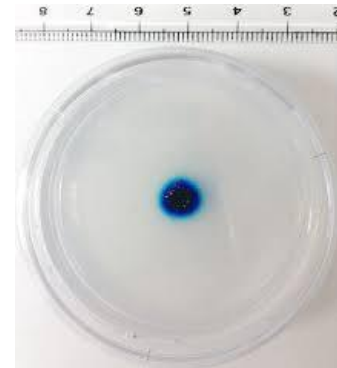
- not rigid
- no fixed shape
- no fixed volume

can be squashed

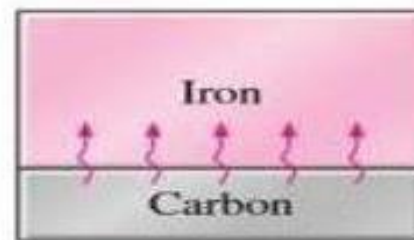
Diffusion Rate

Medium:

gases > liquids > solids



(c) Solid to liquid

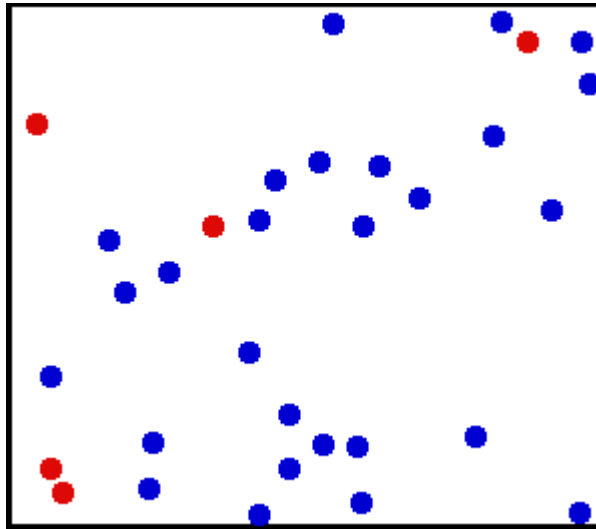


(d) Solid to solid

Diffusion Rate

Temperature:

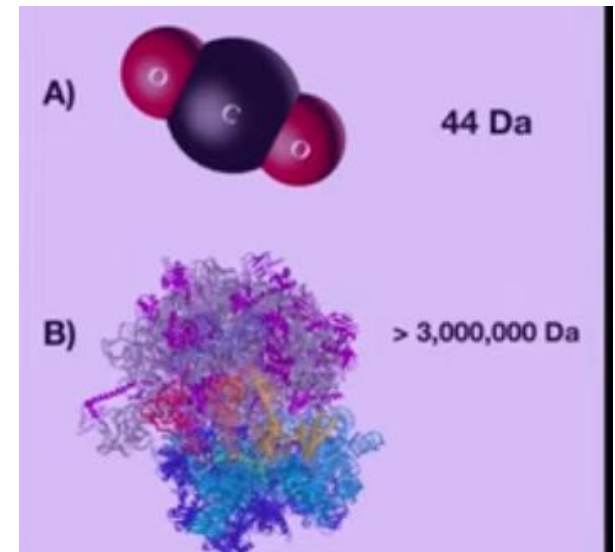
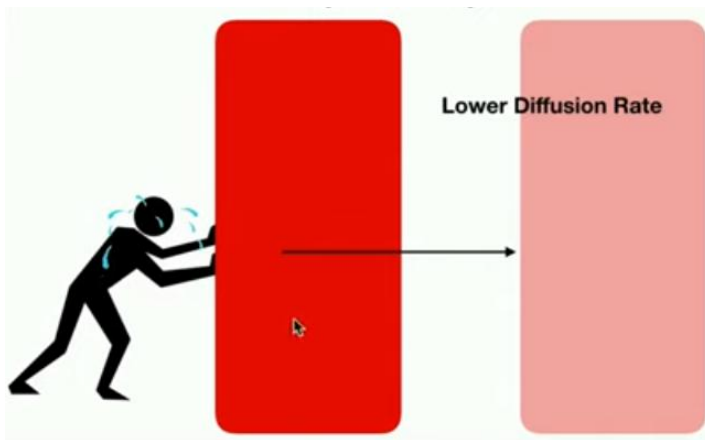
high temperature > low temperature



Diffusion Rate

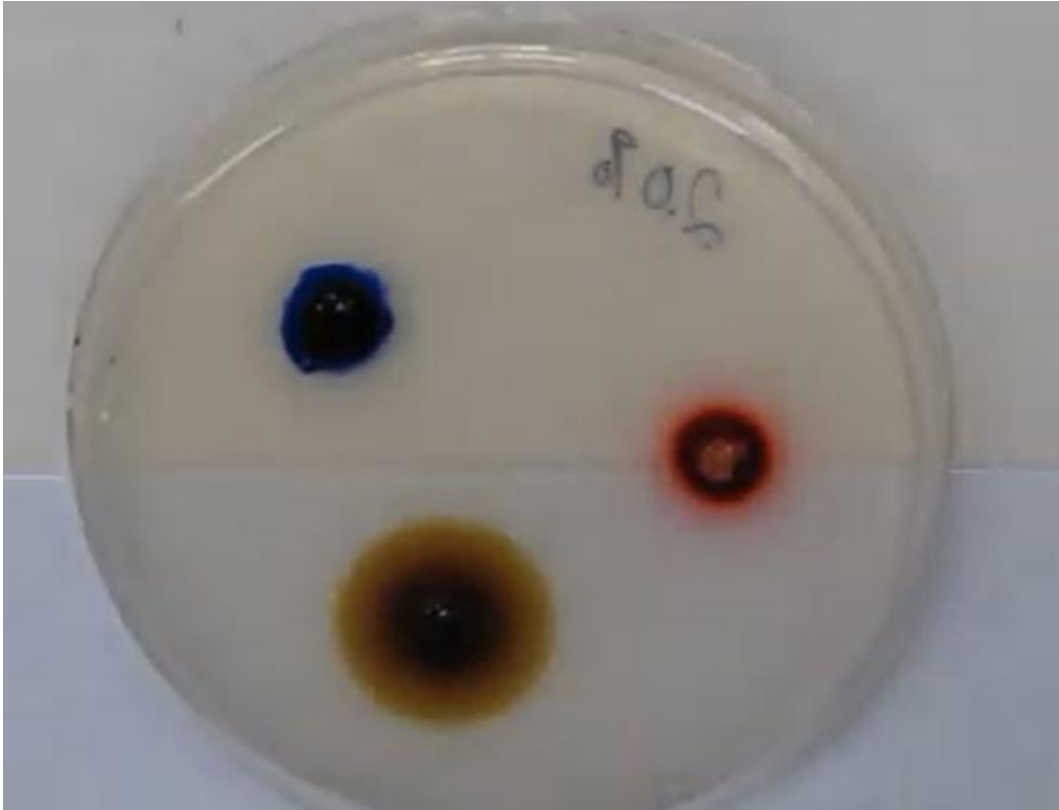
Particle Size/Molecular weight

small particle > large particle



Which one diffuse faster?

Diffusion Rate



Molecular Weights

Safranin - 350.84

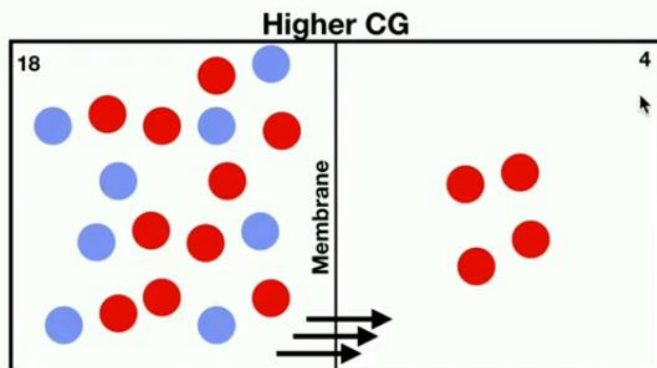
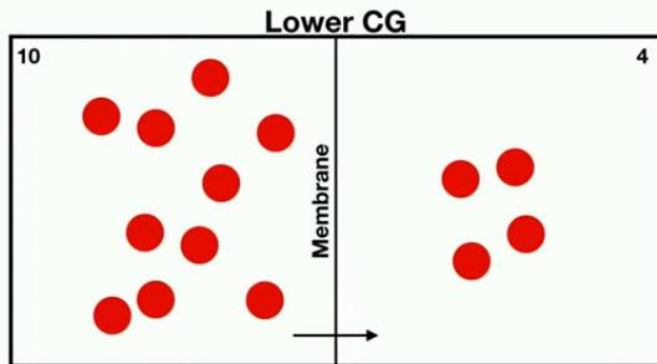
Methylene Blue - 320

Potassium permanganate - 158

Diffusion Rate

Concentration gradient

High concentration gradient > Small concentration gradient

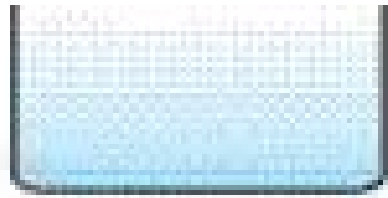


- Concentration gradient:
Concentration difference of a solute
between two points in space

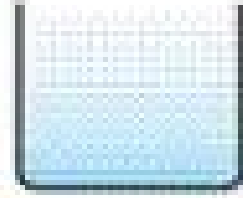
Diffusion Rate

Surface Area

Large SA > Small SA



A

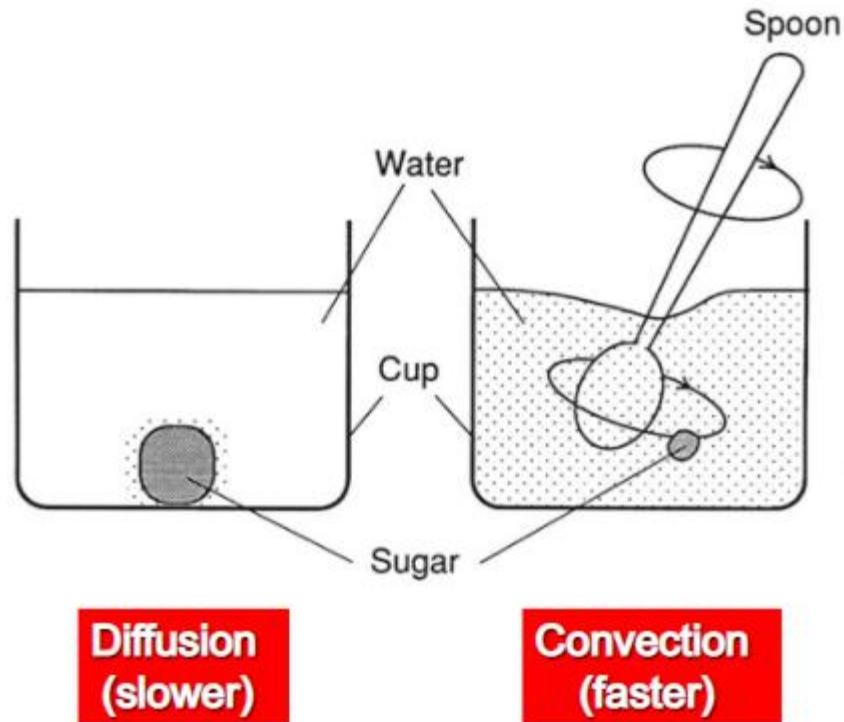


B

Diffusion Rate

Dynamics of system:

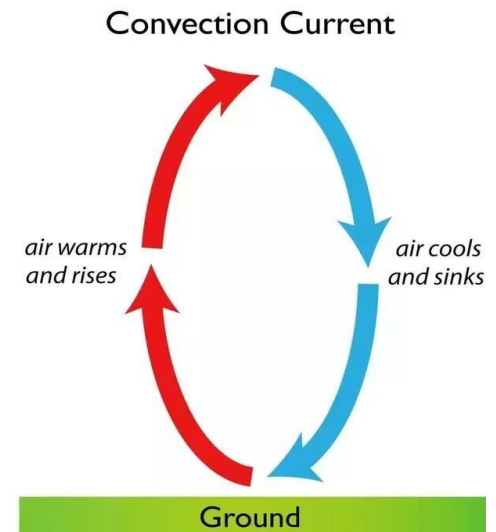
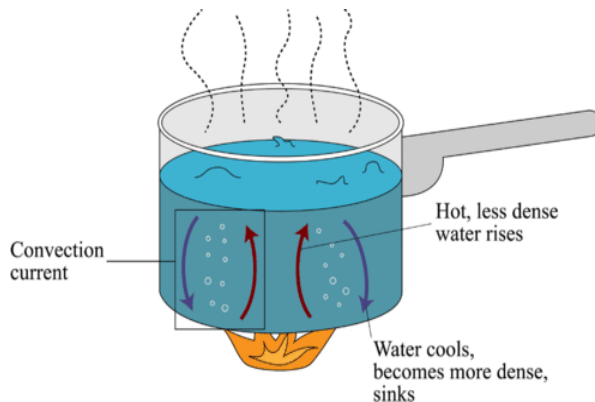
Dynamic > Static



Stirring helps the sugar molecules transfer to the bulk water much faster

Convection

- Mass transfer between a **moving fluid** and a surface or between immiscible moving fluids
- Affected by the **dynamics characteristic** of the flowing fluid
- Fluid motion combines with diffusion to promote the transfer of a species for which there exists a concentration gradient
- Analogous to convection heat transfer

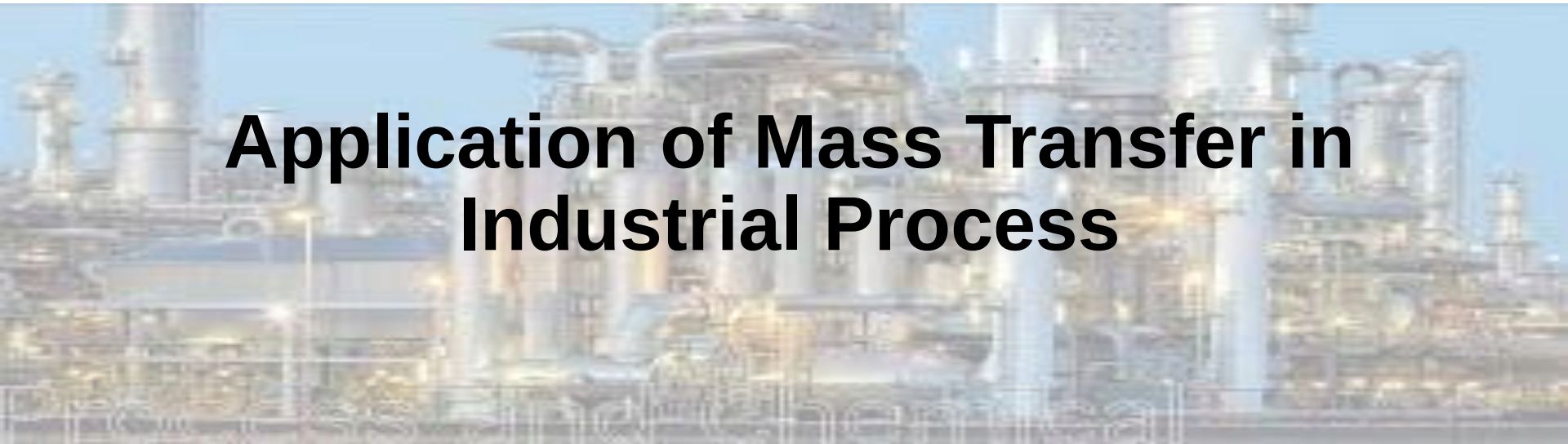


Convection

Example:

- Water evaporates faster in the presence of wind
- Sublimation of naphthalene (solid form) into the moving air
- Spreading of water pollutant in flowing river



A blurred background image of an industrial refinery or chemical plant at night, with various towers, pipes, and lights visible.

Application of Mass Transfer in Industrial Process

Applications

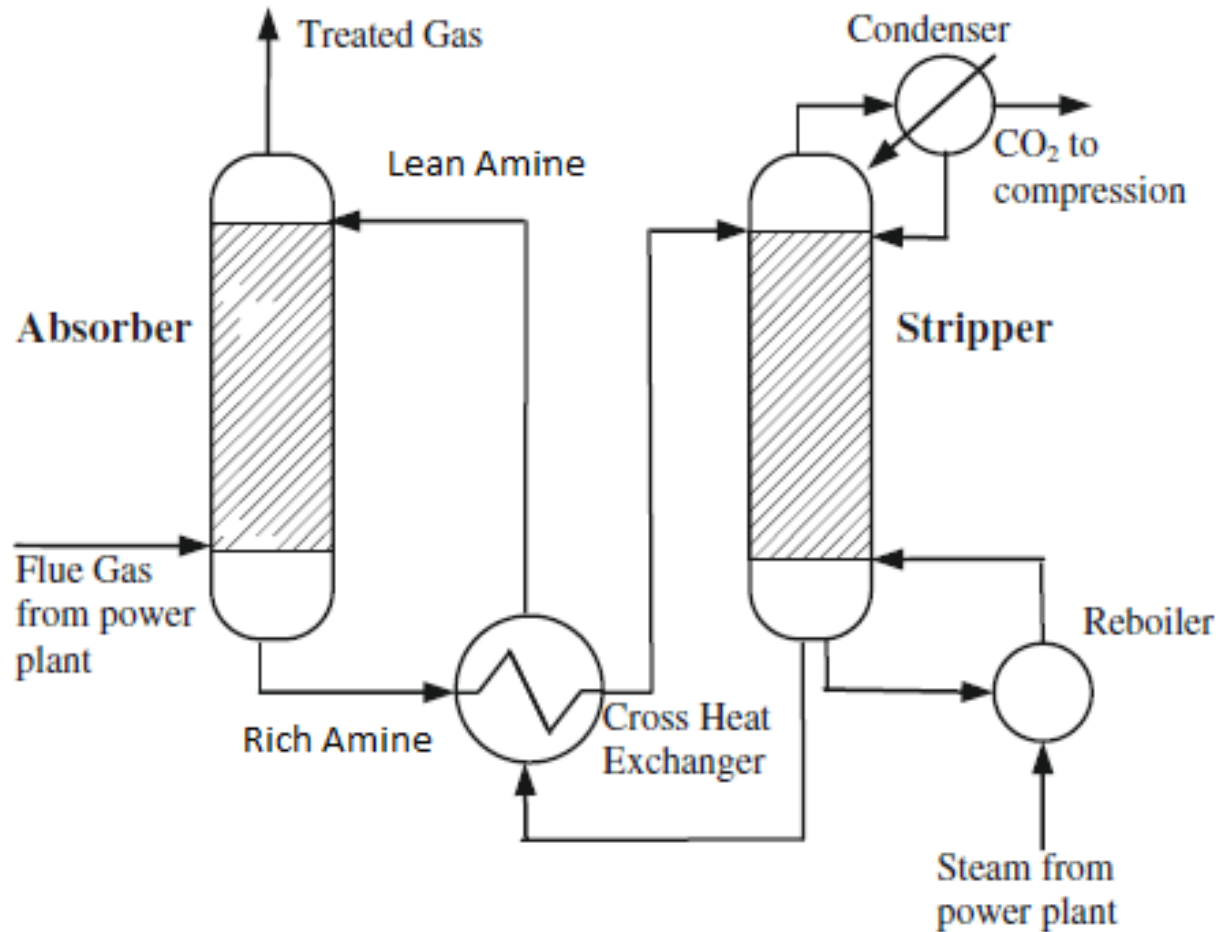
Gas absorption

- Separation of component from gas stream using a solvent
- Example: to capture volatile organic compound, absorption of CO_2 from flue gas

Stripping

- To recover the solvent used in absorption

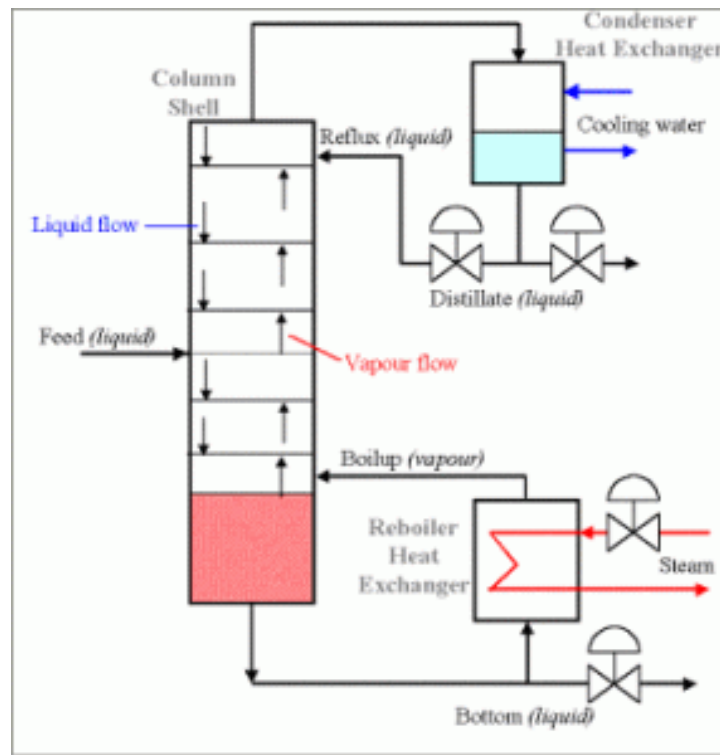
Applications



Applications

Distillation

- Mass transfer between vapor and liquid at each stage



Applications

Liquid-Liquid Extraction

- Separate component based on its relative solubilities in two immiscible solvents
- Net transfer of component from one liquid to another liquid
- Example:
 - Extraction of penicillin from fermentation broth using solvent such as butyl acetate
 - Extraction of acetic acid from water using an organic solvent

Applications

Liquid-Liquid Extraction



Cocurrent



Countercurrent

Applications

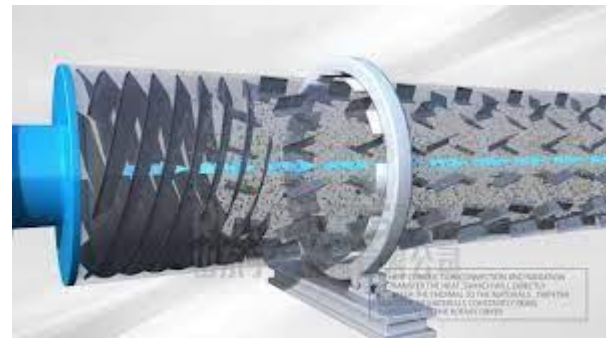
Solid-Liquid Extraction/Leaching

- Or known as solid-liquid extraction
- Extract component out from solid using solvent
- Example:
 - Extraction of caffeine from tea/coffee bean using solvent
 - Oil extraction from soybeans, cotton seeds, peanuts using hexane
 - Extraction of pharmaceutical product from plant roots, leaves, and stems
 - Gold is leached from its ore using aqueous sodium cyanide solution

Applications

Drying

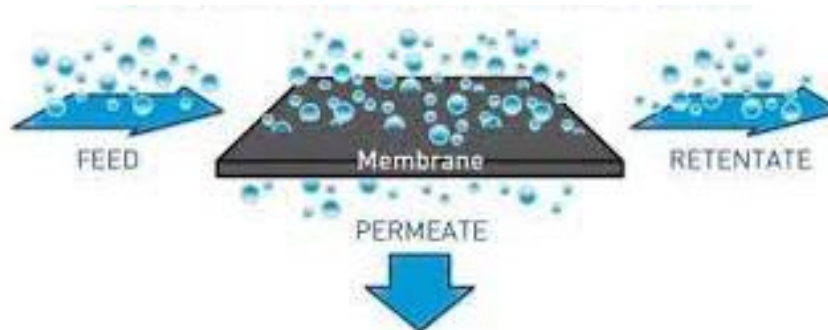
- Dry the material in industry
- Example rotary drum dryer, flash dryer
- Direct (contact with heating stream) or indirect type



Applications

Membrane Separation

- Water separation: Removal of dye, oil, heavy metal contaminant from effluent, produced water
- Gas separation: CO₂ removal, biogas upgrading, hydrogen recovery
- Others: Concentration of product, clarification of fruit juice



Momentum, Energy, and Mass Transfer Analogies

Analogies

Transport Phenomenon	Governing Law	Driving Potential
Mass Transfer	Fick's Law of Diffusion	Concentration Gradient, $\frac{dC_b}{dx}$
Heat Transfer	Fourier's law of conduction	Temperature Gradient, $\frac{dT}{dx}$
Momentum Transfer	Newton's Law of Viscosity	Velocity Gradient, $\frac{du}{dy}$

Analogies

Mass Transfer

$$j = -D \frac{dc}{dx}$$

Heat Transfer

$$q = -k \frac{dT}{dx}$$

Momentum Transfer

$$\tau = -\mu \frac{du}{dy}$$

Shear stress/momentum flux



Analogies

- All three transports can be contributed by **molecular** transport and **bulk flow**
- Mass : Diffusion vs Convection
- Heat: Conduction vs Convection
- Momentum: Viscous vs Convection

Replacement Classes

Original Class	Replacement Class
27 May 2026 (Week 8, Wednesday) 10 am – 12 pm	29 May 2026 (Week 8, Friday) 2 pm – 4 pm TEST
17 June 2026 (Week 11, Wednesday) 10 am – 12 pm	12 June 2026 (Week 10, Friday) 10 am – 12 pm